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Genotypic Distribution and Clinical Correlations in Familial Exudative Vitreoretinopathy: A Single-Center Study

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Purpose: To report the role of wide-angle imaging in detecting suspicious cases of familial exudative vitreoretinopathy (FEVR) in pediatric patients with unexplained vision loss and describe genotypic distribution and examples of phenotypes.

Design: A retrospective cohort study was conducted at a single tertiary referral center in the Intermountain West.

Subjects and Participants: Patients diagnosed with FEVR or atypical retinopathy of prematurity (ROP) between 2010 and 2021 at the University of Utah. Twenty-five families with FEVR were included, with 21 families undergoing genetic testing. Eight families with atypical ROP were included.

Methods: We conducted a retrospective analysis of patients referred with unexplained vision loss and diagnosed with FEVR at the pediatric retina center at the University of Utah from 2010 to 2021. Clinical examination and wide-angle fluorescein angiography (FA) were performed. Patients identified with abnormal peripheral retinal or intravitreal vascularization were recommended for genetic testing. Next-generation sequencing was used to identify variants in known genes associated with FEVR. The positivity rate and the proportion of each positive genetic mutation were calculated. We also include a small cohort of premature infants with atypical ROP who underwent genetic testing prior to the examination under anesthesia.

Main Outcome Measures: Detection rate of FEVR-associated mutations.

Results: Genetic variants were identified in 85.7% of families who underwent testing, exceeding previously reported detection rates. LRP5 (33.3%) and FZD4 (19%) were the most common mutations. Indeterminate results were reported in 4.8% of cases, while 9.5% had negative results for FEVR-associated mutations. Among the 8 premature infants with atypical regression of ROP, none tested positive for FEVR-associated genotypes. We described 5 illustrative cases that demonstrate unique presentations in our cohort, including those showing phenotypic variability or masquerading as other disorders.

Conclusions: The findings highlight the genotypic and phenotypic heterogeneity of FEVR and underscore the value of wide-angle FA to trigger obtaining genetic testing for accurate diagnosis. A high clinical suspicion for FEVR is recommended in pediatric patients with unexplained vision loss and vitreoretinal abnormalities. Future studies are needed to investigate additional genetic modifiers and refine genotype–phenotype correlations.

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Familial exudative vitreoretinopathy (FEVR) is a rare hereditary disorder characterized by incompletely vascularized peripheral retina and abnormal retinal vasculature in full-term subjects and was first described by Criswick and Schepens in 1969. Their report included 6 nonpremature subjects from 2 families with a clinical phenotype similar to retinopathy of prematurity (ROP).[1] Ophthalmoscopic findings of FEVR have since been described to include avascular peripheral retina, exudates, and retinal detachment (RD).[1–4] A clinical staging system based on these clinical features has been reported.[5] However, FEVR is genetically heterogeneous and has variable penetrance, making the clinical appearance among subjects heterogeneous. There are also reported mutations in several genes, including *FZD4*, *LRP5*, *TSPAN12*, and *NDP*, that disrupt the Wnt receptor signaling pathway, which is critical for regulating vascular growth and development, neuronal maturation, and synaptogenesis.[6–8] Mutations in *CTNNB1*, which codes for the protein beta-catenin, the final mediator in Wnt signaling, have also been described previously.[9] *KIF11* belongs to the kinesin family and encodes a spindle motor protein that affects mitosis and has also been associated with an FEVR phenotype.[10] Despite advancements in genetic screening, it is estimated that genetic variants are not identified 50% of the time.[11] In addition, phenotypic expression of mutations varies substantially, complicating diagnosis and management.

This study analyzed the genotypic distribution of patients with clinical FEVR who presented over a 12-year period at

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the University of Utah. Our hypothesis was that the percentage of detected positive genotypes in our cohort would be higher than in previous studies due to the combined use of next-generation sequencing (NGS) and wide-angle imaging for patient selection. We included 2 groups: (1) children, adults, and families with features of FEVR and (2) premature infants with ROP but with delayed regression or features atypical for the natural history of ROP. The second group is based on an emerging classification of overlap between ROP and FEVR, termed ROP vs. FEVR (retinopathy of prematurity versus familial exudative vitreoretinopathy [ROPER]).[†2] Recommendations to consider genetic testing in ROPER are suggested by pediatric retinal specialists, as previously discussed at the Advances in Pediatric Retina meeting in 2023. While the first group allowed for a direct assessment of our hypothesis, the second group was more descriptive due to the limited number of cases. In this article, we also describe 5 illustrative FEVR cases to demonstrate the disease's phenotypic variability and masquerading nature.

Methods

We conducted a retrospective review following the tenets of the Declaration of Helsinki at the University of Utah between 2010 and 2021. The study was approved by the Institutional Review Board at the University of Utah, and informed consent for the study was waived. Inclusion criteria consisted of a diagnosis of FEVR after clinical examination and imaging with wide-angle fluorescein angiography (FA).

Full-term patients were referred with unexplained vision loss and retinal abnormalities to the pediatric retinal center and were evaluated by a pediatric retina specialist (M.E.H.). Familial exudative vitreoretinopathy diagnosis was aided by an examination under anesthesia with wide-angle FA using the handheld RetCam (Clarity Medical Systems) or, in some ages, clinical wide-angle FA with the Optos ultra-widefield retinal imaging device (Optos). Familial exudative vitreoretinopathy was defined as stage $\geq$2,[13] or by 2 of the following findings in $\geq$1 eye: (1) peripheral avascular retina, (2) fluorescein leakage at vascular–avascular junction, and (3) brush-like straightening of the retinal vessels. Genetic consultation and testing were performed based on a suspicion of FEVR after clinical examination and wide-angle FA. Genetic counseling and testing were conducted with the family's consent.

Additionally, we included a group of premature infants who presented with atypical courses of ROP, as defined by deviations from the natural history of ROP described in published studies.[14] These atypical features included slow peripheral retinal vascularization or regression of stage 3 ROP without anti-VEGF treatment. For this subgroup, wide-angle FA was not performed before genetic testing, as the genetic results guided the decision to proceed with an examination under anesthesia.

Next-generation sequencing was performed using one of the following platforms: eyeGENE (National Institutes of Health/National Eye Institute),[15] Invitae (now Labcorp Genetics), Molecular Vision Laboratories, or Blueprint Genetics. These panels included deletion/duplication analysis of genes associated with FEVR, including *FZD4*, *LRP5*, *TSPAN12*, *NDP*, *ZNF408*, *KIF11*, and *CTNNB1*. Reports from eyeGENE provided clinical confirmation of identified pathogenic variants but did not include comprehensive genetic data from the entire panel.

The gene *CTNNB1* was added to FEVR-associated panels after its discovery in 2020. For patients tested before its inclusion, additional testing was offered if no confirmed genetic variant associated with FEVR had been identified in the initial analysis.

Results

We identified 25 families (41 subjects) diagnosed with FEVR (Table 1). Of these, 21 (84%) families underwent genetic testing. Among those tested, 2 families were negative for a genetic variant associated with FEVR. In one of these families (family 11), genetic testing was limited to *FZD4, LRP5,* and *TSPAN12*, whereas in the other family (family 22), the family received a full genetic panel of 266 genes, including *ATOH7*, *CTNBB1*, *FZD4*, *JAG1*, *KIF11*, *LRP5*, *NDP*, *TSPAN12*, and *ZNF408*. Additionally, 1 family had indeterminate results from genetic testing. This yielded a positive mutation identification rate of 85.7% among those tested for FEVR-associated mutations.

Of the 21 families that underwent genetic testing, the proportions of identified mutations were as follows: *LRP5* (33.3%), *FZD4* (19.0%), *TSPAN12* (9.5%), *KIF11* (9.5%), *CTNNB1* (9.5%), and *NDP* (4.8%) (Fig 1). Indeterminate results were reported in 4.8% of cases, while 9.5% had negative results for FEVR-associated mutations.

Among the 8 premature infants with atypical regression of ROP, none tested positive for FEVR-associated genotypes, including *CTNNB1*, which was tested in all cases (Table 2).

Illustrative Cases

To highlight the phenotypic diversity of FEVR, we present 5 illustrative cases selected from our cohort. These cases were not chosen to represent the entire cohort but rather to demonstrate the wide variability in clinical presentation and the potential for misdiagnosis in pediatric patients. Familial exudative vitreoretinopathy can present at different ages with a range of severity, sometimes mimicking other diseases, as seen in case 1 (retinoblastoma concern) and case 4 (atypical vitreous hemorrhage with hydrocephalus). These examples underscore the importance of maintaining high clinical suspicion for FEVR, particularly in children with unexplained vision loss or retinal findings.

Case 1: Vitreous Hemorrhage with Vasoproliferative Tumor

Description. A 6-year-old male (subject 24) presented with vitreous hemorrhage in the right eye (OD), visual acuity of 20/100 OD, and 20/40 in the left eye (OS). There was no view of the fundus using indirect ophthalmoscopy. By ultrasonography, there was a concern for retinoblastoma based on the finding of a peripheral mass with high internal reflectivity on B-scan ultrasound (Fig 2A). A second opinion from ocular oncology was received, which weighed against retinoblastoma. In addition, OS showed peripheral avascular retina with abnormal vascularization at the junction of vascular/avascular retina and exudates that were suggestive of FEVR (Fig 2B, C). The patient underwent a vitrectomy OD with careful dissection of

Table 1. Genetic Testing of Families with FEVR

Family Identifier	Subject Identifiers	Genetic Testing	Lab	Test	Result	Gene	Variant(s)	Classification
1	1	Yes	EyeGene		Positive	*FZD4*	c.1282_1285del, p.Asp428Serfs*2	Pathogenic
2	2, 3, 4	Yes	Blueprint Genetics	IRD panel	Positive	*LRP5*	c.2718_2721del, p.Met907Thrfs*52	Pathogenic
							c.3232C > T, p.Arg1078*	Pathogenic
3	5	No						
4	6, 7	Yes	Connective Tissue Gene Tests	Vitreoretinopathy panel	Positive	*LRP5*	c.632G > A, p.Trp211	Pathogenic
5	8	Yes	Invitae	IRD panel	Positive	*KIF11*	Deletion of exon 10	Pathogenic
6	9	No						
7	10	Yes	Invitae	Vitreoretinopathy panel	Positive	*FZD4*	Deletion of entire coding sequence	Pathogenic
8	11	Yes	EyeGene		Positive	*FZD4*	c.313A > G, p.Met105Val	Pathogenic
9	12	Yes	Molecular Vision Laboratories	FEVR panel	Positive	*LRP5*	c.3127G > T, p.Glu1043*	Likely pathogenic
10	13	Yes	EyeGene		Positive	*LRP5*	c.1169C > G, p.Thr390Arg	Likely pathogenic
11	14	Yes	University of Utah Lab	FEVR panel*	Negative			
12	15	Yes	Invitae	IRD panel†	Indeterminate	*BEST1*	c.290A > G, p.Tyr97Cys	Variant of uncertain significance
13	16, 17	Yes	EyeGene		Positive	*LRP5*	c.1879T > C, p.Cys627Arg	Likely pathogenic
14	18	Yes	Blueprint Genetics	Vitreoretinopathy panel	Positive	*CTNNB1*	c.1982G > C, p.(Arg661Pro)	Likely pathogenic
15	19, 20, 21, 22	Yes	EyeGene		Positive	*TSPAN12*	c.302T > A, p.Leu101His	Likely pathogenic
16	23, 24, 25	Yes	Blueprint Genetics	Vitreoretinopathy panel	Positive	*TSPAN12*	c.41T > G, p.Leu14Arg	Likely pathogenic
17	26	Yes	Invitae	IRD panel	Positive	*KIF11*	c.388-2A > T	Pathogenic
18	27	Yes	Invitae	Vitreoretinopathy panel	Positive	*CTNNB1*	c.1723G > A, p.Gly575Arg	Likely pathogenic
19	28	Yes	Invitae	Vitreoretinopathy panel	Positive	*NDP*	c.243C > G (p.Phe81Leu)	Likely pathogenic
20	29	Yes	Invitae	IRD panel	Positive	*LRP5*	c.2737dup, p.Cys913Leufs*73	Pathogenic
21	30, 31	Yes	EyeGene		Positive	*FZD4*	c.617dup, p.Tyr206*	Likely pathogenic
22	32	Yes	Blueprint Genetics	IRD panel	Negative			
23	33	No						
24	34	No						
25	35, 36, 37, 38, 39, 40, 41	Yes	Invitae	IRD panel	Positive	*LRP5*	c.4488+2T > G	Pathogenic

FEVR = familial exudative vitreoretinopathy; IRD = inherited retinal disorder.
*Only FZD4, LRP5, and TSPAN12 were tested.
†CTNNB1 not tested.

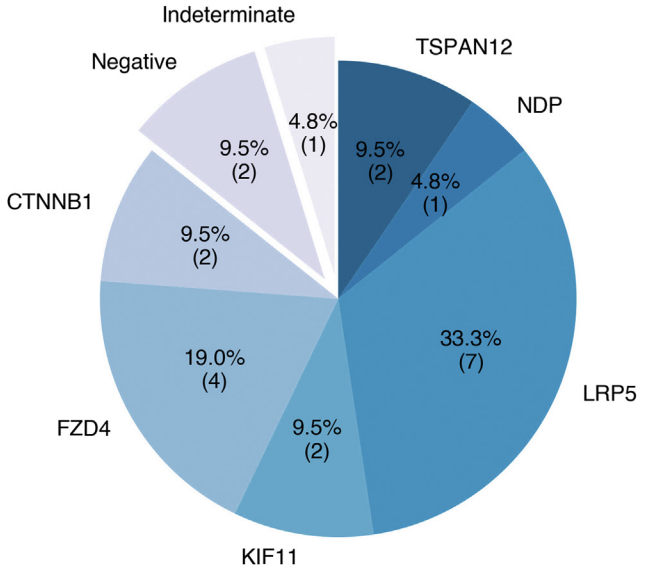


Figure 1. Summary of genetic testing results in families who underwent genetic testing. The genetic positivity rate in our cohort was 85.7%. Numbers in parentheses indicate the number of families for that group.

vitreoretinal membranes and laser to the peripheral avascular retina (Fig 2D, E). There was an elevated lesion noted with adherent blood that was not safely removed. After clearance of the blood in postoperative examinations, a diagnosis of a vasoproliferative tumor was made. The tumor was treated with cryotherapy about 3 months after the vitrectomy. Visual acuity improved to 20/60 OD and 20/25+ OS.

Genotype. *TSPAN12*, c.41T>G, p.Leu14Arg.

Discussion. This case highlights the diverse clinical presentations of FEVR, in which a patient with a TSPAN12 mutation presented with a vasoproliferative tumor and vitreous hemorrhage. The vasoproliferative tumor has been reported in FEVR,[16] but in this case, it led to the concern of a potential retinoblastoma, necessitating comprehensive diagnostic evaluations to ensure accurate diagnosis and appropriate management.

Case 2: Variability between Eyes and within Families with the Same Genotype

Description. A 12-year-old girl (subject 1) presented with an epiretinal membrane and peripheral traction OS (Fig 3A, B). Visual acuity OD was 20/50, and OS was 20/60. Scatter laser photocoagulation was also performed on the peripheral avascular retina appreciated by wide-angle FA. The patient was followed until 6 months later when the visual acuity dropped to 20/100 OS from a tractional macular detachment. The patient underwent a lens-sparing vitrectomy and epiretinal membrane removal OS with segmentation of membranes near the vasculature that restored vision OS to 20/60 (Fig 3C, D).

Her younger 5-year-old sister presented with a rhegmatogenous RD involving the macula and lattice degeneration with holes in the avascular peripheral retina. Her visual

Table 2. Genetic Testing in Families with ROP vs. FEVR (ROPER)

Family Identifier	Subject Identifiers	Genetic Testing	Lab	Test	Result	Gene	Variant(s)	Classification
32	42	Yes	Invitae	Vitreoretinopathy panel	Negative			
33	43	Yes	Invitae	Vitreoretinopathy panel*	Negative			
34	44	Yes	Invitae	Vitreoretinopathy panel	Negative			
			ARUP Laboratories	Cytogenomic SNP Microarray	Indeterminate	16p13.11	Duplication	Variant of uncertain significance
35	45	Yes	Blueprint Genetics	Vitreoretinopathy panel	Negative			
36	46	Yes	Invitae	IRD panel*	Negative			
37	47	Yes	Invitae	Vitreoretinopathy panel	Negative			
38	48	Yes	Invitae	Vitreoretinopathy panel*	Negative			
39	49	Yes	Invitae	IRD panel*	Negative			

FEVR = familial exudative vitreoretinopathy; IRD = inherited retinal disorder; ROP = retinopathy of prematurity.
*CTNNB1 not tested.

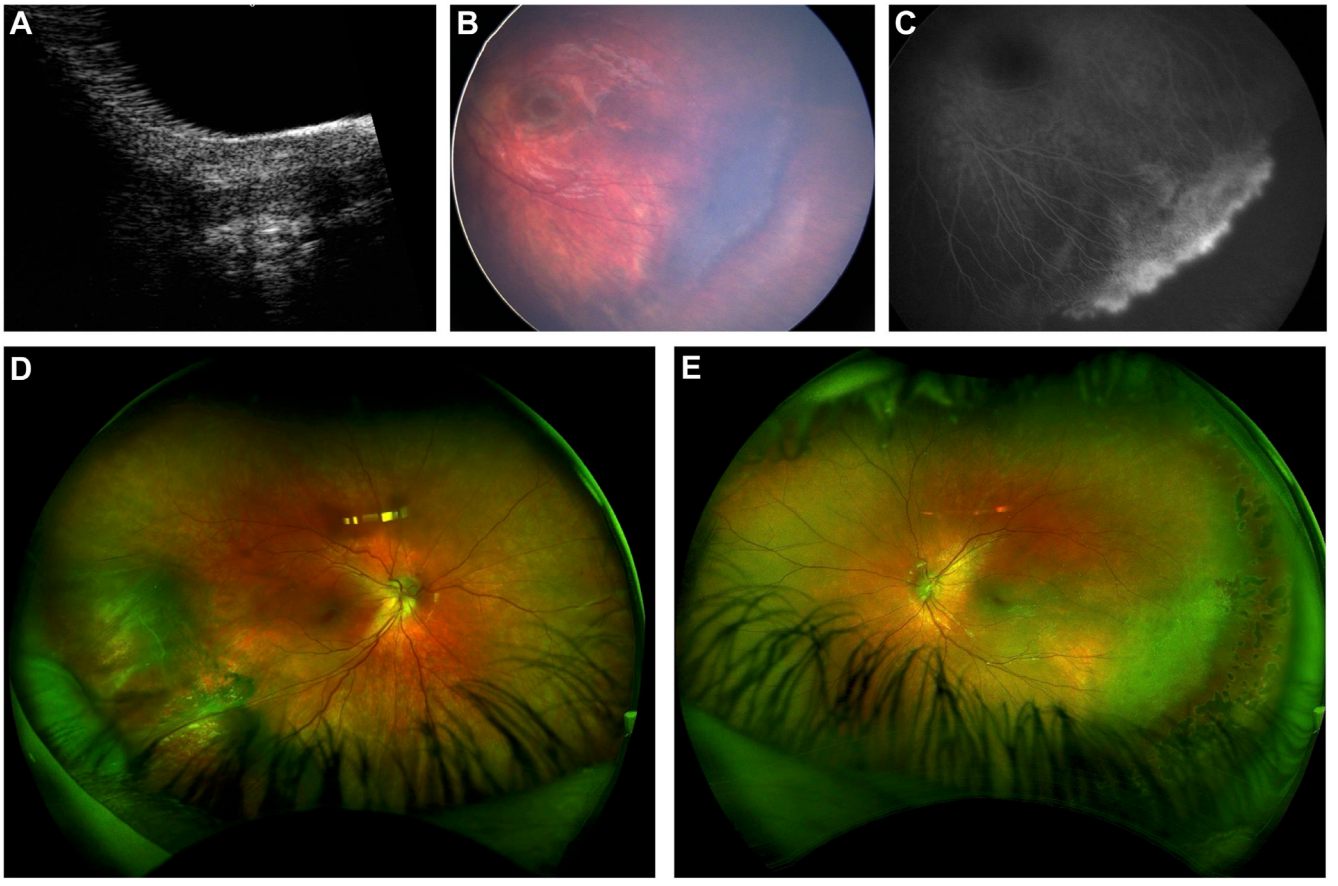


Figure 2. Case 1. **A,** B-scan ultrasound of a peripheral mass with high reflectivity in the right eye. **B,** Avascular peripheral retina in the left eye. **C,** Fluorescein angiography demonstrating abnormal vascularization at the vascular and avascular junction. **D,** Ultra-widefield Optos image of the right eye with a treated inferotemporal vasoproliferative tumor after pars plana vitrectomy. **E,** The left eye had an avascular retina treated with laser photocoagulation.

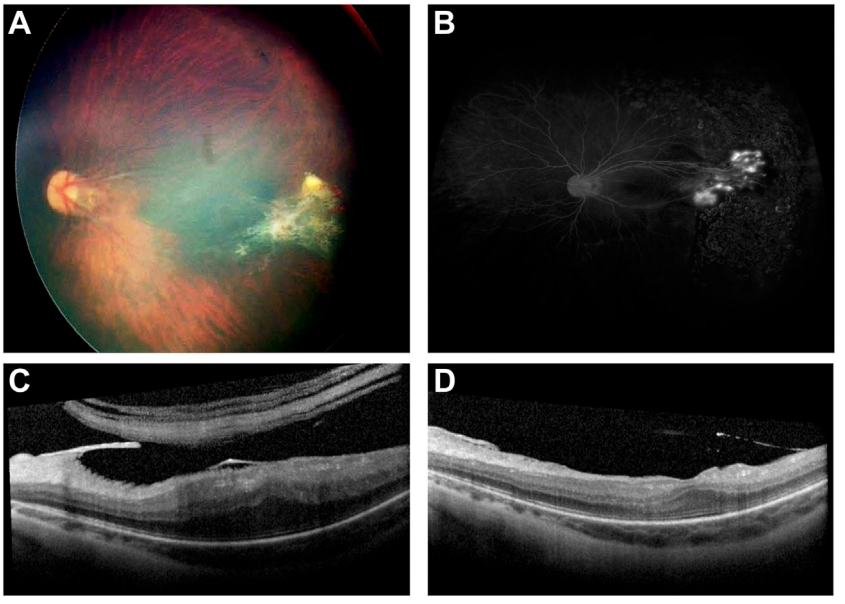


Figure 3. Case 2. **A,** Retcam photograph of the left eye of the proband, demonstrating dragging of the retinal vessels, ERM, and peripheral traction. **B,** Fluorescein angiogram after scatter photocoagulation. **C,** Preoperative OCT demonstrating ERM and traction in the macula of the left eye. **D,** Postoperative OCT after pars plana vitrectomy with ERM removal. ERM = epiretinal membrane.

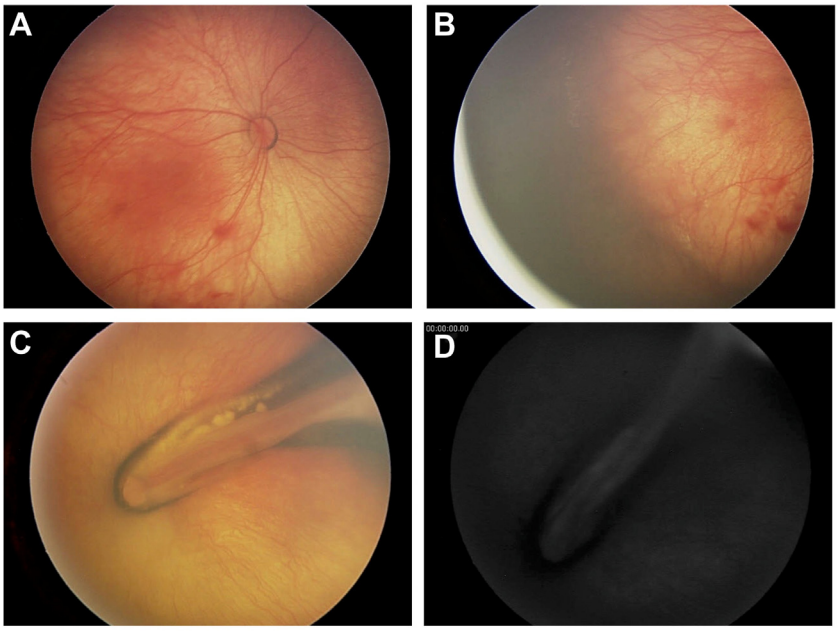


Figure 4. Case 3. **A,** Fundus photograph of the right eye with areas of intraretinal hemorrhages. **B,** Peripheral retina with intraretinal hemorrhages abutting avascular retina. **C,** Fundus photograph of the left eye with a large retinal fold. **D,** Fluorescein angiography of the left eye shows avascular peripheral retina because the retinal vasculature was drawn up into the fold of retina.

acuity was 20/500 in the affected eye (OD) and 20/20 in the unaffected eye (OS). Her OS showed areas of peripheral avascular retina. She underwent scleral buckle and cryotherapy with drainage of subretinal fluid that reattached her retina and maintained visual acuity.

Genotype. *FZD4*, c.617dup, p.Tyr206*.

Discussion. This family underscores the phenotypic variability both between eyes in the same patient and between siblings with the same genotype. While the older sister showed peripheral traction and an epiretinal membrane, the younger sister had a rhegmatogenous RD.

Case 3: Variability in Same Patient

Description. A full-term 3-month-old girl (subject 13) was referred for intraretinal hemorrhages OD and an unusual appearance OS. Fundus photography OD revealed intraretinal hemorrhages in the inferior macula and in the temporal periphery OD (Fig 4A, B). Photography OS revealed a large retinal fold extending from the optic nerve toward the temporal periphery (Fig 4C, D). The patient had no known family history of FEVR and was not premature. Genetic testing revealed an *LRP5* mutation. Interestingly, the patient's mother, who also carried the *LRP5* mutation, had a normal retinal examination except for clumped pigmentation. There was no peripheral avascular retina in the mother on FA.

Genotype. *LRP5*, c.1169C>G, p.Thr390Arg.

Discussion. This case illustrates the variability of FEVR even within the same individual and across generations. The significant difference in phenotypic expression between the mother and daughter with the same *LRP5* mutation highlights the complex nature of genotype–phenotype correlations in FEVR.

Case 4: Vitreous Hemorrhage of Unknown Etiology

Description. A 20-year-old female (subject 27) with spina bifida and hydrocephalus treated with a ventriculoperitoneal shunt presented with a vitreous hemorrhage OD. The OS was entirely normal. Visual acuity was 20/200 OD and 20/40 OS. Initially, there was concern that the intracranial pressure might be elevated due to shunt malfunction. However, testing for the shunt and intracranial pressure was normal. A vitrectomy revealed avascular peripheral retina consistent with stage 2 FEVR (Fig 5A). The OS was normal (Fig 5B). Visual acuity improved to 20/30 in both eyes.

Genotype. *CTNNB1*, c.1723G>A, p.Gly575Arg.

Discussion. This case highlights the differential diagnosis for vitreous hemorrhage in children where FEVR should always be considered. The complex medical history of a ventriculoperitoneal shunt led to other initial differential diagnoses for the vitreous hemorrhage.

Case 5: Progression with Childhood Growth

Description. A 10-year-old female (subject 22) presented with dragging of the macula and avascular peripheral retina OD and fibrosis with peripheral avascular retina OS. She was diagnosed with FEVR. She and several family members were found to have a variant in *TSPAN12*. The patient underwent scatter laser photocoagulation in both eyes (Fig 6A). The retinas appeared stable for 5 years. Exudation

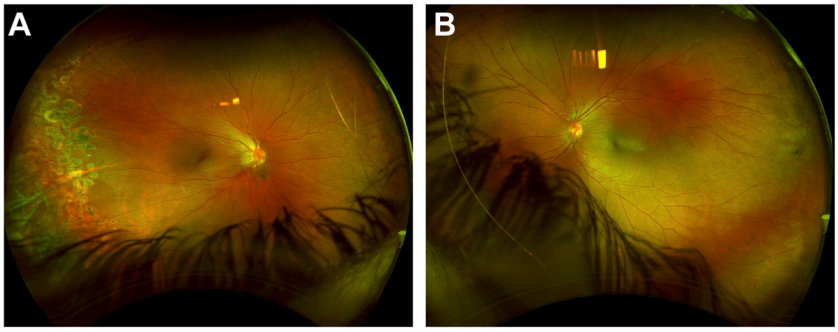


Figure 5. Case 4. **A,** Ultra-widefield Optos photo of the right eye after vitrectomy and scatter photocoagulation to avascular peripheral retina. **B,** Normal ultra-widefield Optos photo of the left eye.

then occurred and was observed at age 12 (Fig 6B), requiring scleral buckling and vitrectomy.

Genotype. *TSPAN12*, c.302T>A,p.Leu1001His.

Discussion. Despite intervention, significant exudation was noted at the age of 12 years, underscoring the dynamic nature of FEVR and the need for close long-term monitoring. Subsequent treatment strategies may involve vitrectomy if tractional components are identified. Pharmaceutical agents, such as anti-VEGF, may also be considered, although the evidence for their efficacy in FEVR remains limited.[17] In children, it can be challenging to see peripheral avascular retina due to limited cooperation with retinal examinations in the clinic or when the view is obscured. In cases with suspicion, wide-angle FA is helpful. Optos ultra-widefield imaging may be performed in the clinic without initially needing anesthesia.

Discussion

Familial exudative vitreoretinopathy is a clinical diagnosis of peripheral avascular retina in full-term infants with combinations of extraretinal neovascularization, bleeding, tractional RDs, and exudative RDs. Although previous studies had genetic confirmation of a variant in <50% of patients, our study confirmed a genetic variant in 85.7% of families that underwent genetic testing when the sample of patients tested had wide-angle FA suspicious for FEVR features. We speculate that the higher positive yield results from identifying patients with suspicious vasculature prior to NGS testing panels that encompass most of the known associated variants with FEVR. In addition, more recently discovered variants, such as *CTNNB1*, are included in such panels. Despite wide-angle FA and NGS testing, we still had 2 families with negative test results. One family (family 11) underwent limited testing for only *FZD4, LRP5,* and *TSPAN12,* performed at the University of Utah. In this case, we suspect a larger NGS panel could have found the causative genetic mutation. In the other family (family 22), a panel test of 266 genes was performed, and the results were negative. In this case, the causative variant could be a genetic variant in more recently associated genes, such as *DOCK6, ARHGAP31,* and *SNX31*, which were not included in the panel.[18,19] We obtain genetic testing for our patients with suspected FEVR after comprehensive examination and imaging with wide-angle FA for multiple reasons, such as support for the clinical diagnosis, information about potential-associated systemic diseases, and family counseling. Although FEVR is a clinical diagnosis, when a genetic mutation related to FEVR is found in a suspected patient, there is strong confidence that the diagnosis is correct.

Several studies have attempted to establish genotype−phenotype correlations in FEVR, with varying degrees of success.[20–22] For example, Wang et al[21] analyzed the clinical and genetic characteristics of retinal folds in 89 patients. Retinal folds were bilateral in 41.6% of patients, predominantly located temporally, and

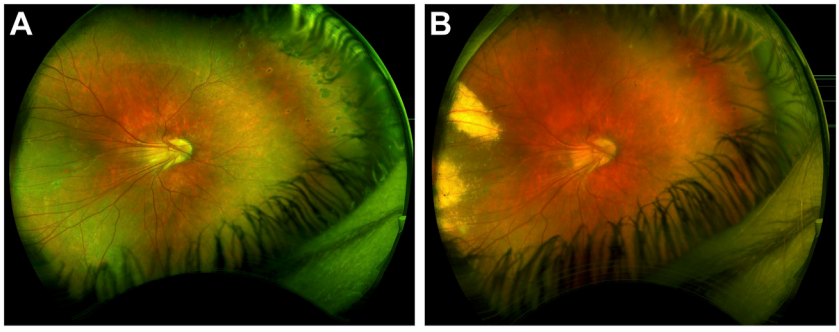


Figure 6. Case 5. **A,** Ultra-widefield Optos photo of the right eye at age 10. **B,** Ultra-widefield Optos photo of the right eye at age 12 with exudation in the periphery.

complete in 97.6% of cases. Genetic testing found variants associated with FEVR in 67.5% of probands. Patients with *LRP5* and *FZD4* mutations had unilateral folds, whereas those with *NDP*, *TSPAN12*, and *KIF11* mutations had bilateral symmetrical severe retinopathy. In our cohort, we had a similar case (case 3) with a unilateral fold and an *LRP5* mutation. Similarly, Tsai et al[22] reviewed clinical outcomes in 40 FEVR patients and found that genotype–positive patients tended to have more severe disease manifestations. A large meta-analysis by Wang et al,[23] which included 3257 patients across 32 studies, identified *NDP* mutations as the most strongly associated with severe disease, followed by *FZD4*, *LRP5*, and *TSPAN12*. However, despite these efforts, genotype–phenotype correlations remain inconsistent, and the clinical spectrum within individual genotypes remains highly variable. Our findings support this complexity, as even within our limited cohort, we observed substantial phenotypic variability among patients with the same mutation. Some of this variability may be due to gene–gene or epigenetic interactions as well as the evolution of pathophysiology that occurs with ischemia in avascular retina or growth spurts identified in children, as examples.

Despite the possibility of prognostication with genetic testing, there is still considerable phenotypic variability in cases of FEVR. We provided examples of variability within families and the same patients (cases 2 and 3). Also, we provided examples in which the eye manifestations were not diagnosed initially as FEVR (cases 1 and 4) to emphasize the importance of having FEVR in the differential diagnosis, particularly in conditions affecting children. It is hypothesized that the various phenotypes in FEVR can be attributed in part to the degree of norrin/β-catenin signal transduction shown by functional assays.[24] For example, a nonsense mutation in *FZD4* may completely abolish signaling, while missense mutations in *LRP5* and *FZD4* may cause a moderate level of reduction.[25] Further studies are required to investigate the molecular underpinnings of the phenotypic variability in FEVR.

Since FEVR is hereditary, parents of affected children may also carry mutations and be at risk of vision loss. Comprehensive family screening and genetic counseling are essential to identify asymptomatic carriers and manage their condition proactively. Kashani et al[26] examined the prevalence and severity of FEVR in asymptomatic relatives of symptomatic FEVR patients. A retrospective chart review of 74 subjects (148 eyes) from 17 families was conducted at a single tertiary referral vitreoretinal practice. The cohort included 17 symptomatic patients and 57 asymptomatic relatives, 43% with detectable mutations in *FZD4, NDP,* or *TSPAN12*. Only 8% reported a positive family history of FEVR. Among symptomatic patients, 76% had advanced stages (3, 4, or 5) of FEVR, while 58% of asymptomatic relatives had early-stage (1 or 2) clinical or angiographic findings. The study concluded that asymptomatic family members often exhibit early signs of FEVR, which can progress to more severe stages, and emphasized the importance of screening and clinical examination for immediate relatives of symptomatic FEVR patients. Even the presence of peripheral avascular retina can predispose to breaks and rhegmatogenous RDs, which may have occurred in the young sister of case 2.

We routinely perform genetic testing for infants exhibiting delayed vascularization in infants with ROP of older postmenstrual ages. In our study, none of the 8 premature infants with atypical regression of ROP tested positive for FEVR-associated genetic variants. This finding highlights the importance of distinguishing between atypical ROP and FEVR, particularly when delayed vascularization or regression patterns deviate from the natural history of ROP. John et al[12] proposed a classification termed ROPER (ROP vs. FEVR) for patients that straddle the distinct entities of ROP and FEVR. While the ROPER framework is valuable, our findings suggest that not all cases of atypical ROP are genetically linked to FEVR. Given our small sample size in this subgroup, our findings are primarily descriptive, and more extensive studies are needed to better define the genetic and clinical boundaries between ROP and FEVR. We believe that identifying any FEVR-related mutations in these patients can provide crucial insights into the potential overlap between FEVR and ROP and inform treatment decisions, especially the need for close monitoring and treatment of peripheral avascular retina when indicated.

A primary limitation of this study is its single-center, retrospective design, which may reduce the generalizability of the findings to other clinical settings and patient populations. However, our study does capture a population in a large area of the Mountain West region of the United States—an area that has not been previously examined for FEVR in the US cohorts.[27,28] Additionally, variable genetic panels were used over the 12-year study period, and genes such as *CTNNB1* were added more recently, possibly underestimating rarer FEVR-associated variants in older tests. In cases where no mutation was initially found, families were offered repeat genetic testing once newer gene panels became available, partially mitigating this limitation.

Our study's strengths include comprehensive genetic testing panels incorporating newly identified FEVR-related genes, leading to a high mutation identification rate. The integration of ultra-widefield imaging enhanced phenotypic characterization and helped distinguish FEVR from similar retinal pathologies, as demonstrated by our illustrative case examples, and increased the genetic positivity of clinical FEVR or suspected cases. Moreover, including both FEVR and ROPER cases broadened our understanding of the disease spectrum. These strengths underscore the importance of pairing advanced imaging with targeted genetic testing to refine diagnostics, enable early intervention, and guide personalized management strategies in FEVR. Our study suggests that future studies, with a greater subject count and a similar combination of ultra-widefield imaging and next-generation genotype sequencing, are warranted to further characterize the landscape of genotype and phenotype of patients with FEVR.

Footnotes and Disclosures

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HUMAN SUBJECTS:

Human subjects were included in this study. We conducted a retrospective review following the tenets of the Declaration of Helsinki at the University of Utah between 2010 and 2021. The study was approved by the Institutional Review Board at the University of Utah, and informed consent for the study was waived.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Soetikno, Hartnett

Data collection: Spoth

Analysis and interpretation: Soetikno, Hartnett

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Overall responsibility: Soetikno, Spoth, Hartnett

Abbreviations and Acronyms:
FA = fluorescein angiography; **FEVR** = familial exudative vitreoretinopathy; **NGS** = next-generation sequencing; **OD** = right eye; **OS** = left eye; **ROP** = retinopathy of prematurity; **ROPER** = retinopathy of prematurity versus familial exudative vitreoretinopathy; **RD** = retinal detachment.

Keywords:
Familial exudative vitreoretinopathy, Genotype, Sequencing.

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References

1. Criswick VG, Schepens CL. Familial exudative vitreoretinopathy. *Am J Ophthalmol*. 1969;68:578–594.
2. Sızmaz S, Yonekawa Y, M TT. Familial exudative vitreoretinopathy. *Turk J Ophthalmol*. 2015;45:164–168.
3. Tauqeer Z, Yonekawa Y. Familial exudative vitreoretinopathy: pathophysiology, diagnosis, and management. *Asia Pac J Ophthalmol*. 2018;7:176–182.
4. Gilmour DF. Familial exudative vitreoretinopathy and related retinopathies. *Eye*. 2015;29:1–14.
5. Ranchod TM, Ho LY, Drenser KA, et al. Clinical presentation of familial exudative vitreoretinopathy. *Ophthalmology*. 2011;118:2070–2075.
6. Rosso SB, Inestrosa NC. WNT signaling in neuronal maturation and synaptogenesis. *Front Cell Neurosci*. 2013;7:103.
7. Xu Q, Wang Y, Dabdoub A, et al. Vascular development in the retina and inner ear: control by norrin and frizzled-4, a high-affinity ligand-receptor pair. *Cell*. 2004;116(6):883–895.
8. Wang Z, Liu C-H, Huang S, Chen J. Wnt Signaling in vascular eye diseases. *Prog Retin Eye Res*. 2019;70:110–133.
9. Dixon MW, Stem MS, Schuette JL, et al. CTNNB1 mutation associated with familial exudative vitreoretinopathy (FEVR) phenotype. *Ophthalmic Genet*. 2016;37:468–470.
10. Hu H, Xiao X, Li S, et al. KIF11 mutations are a common cause of autosomal dominant familial exudative vitreoretinopathy. *Br J Ophthalmol*. 2016;100:278–283.
11. Rao F-Q, Cai X-B, Cheng F-F, et al. Mutations in LRP5, FZD4, TSPAN12, NDP, ZNF408, or KIF11 genes account for 38.7% of Chinese patients with familial exudative vitreoretinopathy. *Invest Ophthalmol Vis Sci*. 2017;58:2623–2629.
12. John VJ, McClintic JI, Hess DJ, Berrocal AM. Retinopathy of prematurity versus familial exudative vitreoretinopathy: report on clinical and angiographic findings. *Ophthalmic Surg Lasers Imaging Retina*. 2016;47:14–19.
13. Pendergast SD, Trese MT. Familial exudative vitreoretinopathy. Results of surgical management. *Ophthalmology*. 1998;105:1015–1023.
14. Schaffer DB, Palmer EA, Plotsky DF, et al. Prognostic factors in the natural course of retinopathy of prematurity. The cryotherapy for retinopathy of prematurity cooperative group. *Ophthalmology*. 1993;100:230–237.
15. Goetz KE, Reeves MJ, Tumminia SJ, Brooks BP. eyeGENE(R): a novel approach to combine clinical testing and researching genetic ocular disease. *Curr Opin Ophthalmol*. 2012;23:355–363.
16. Shields CL, Kaliki S, Al-Dahmash S, et al. Retinal vasoproliferative tumors: comparative clinical features of primary vs secondary tumors in 334 cases. *JAMA Ophthalmol*. 2013;131:328–334.
17. Lyu J, Zhang Q, Xu Y, et al. Intravitreal ranibizumab treatment for advanced familial exudative vitreoretinopathy with high vascular activity. *Retina*. 2021;41:1976–1985.
18. Tao Z, Bu S, Lu F. Two AOS genes attributed to familial exudative vitreoretinopathy with microcephaly: two case reports. *Medicine (Baltimore)*. 2021;100:e24633.
19. Xu N, Cai Y, Li J, et al. An SNX31 variant underlies dominant familial exudative vitreoretinopathy-like pathogenesis. *JCI Insight*. 2023;8:e167032.
20. Kondo H, Tsukahara-Kawamura T, Matsushita I, et al. Familial exudative vitreoretinopathy with and without pathogenic variants of norrin/β-catenin signaling genes. *Ophthalmol Sci*. 2024;4:100514.
21. Wang Z, Chen C, Sun L, et al. Symmetry of folds in FEVR: a genotype-phenotype correlation study. *Exp Eye Res*. 2019;186:107720.
22. Tsai ASH, Kang EYC, Wang NK, et al. Long-term clinical outcomes and genotype-phenotype correlation in familial exudative vitreoretinopathy in a tertiary referral center. *Retina*. 2023;43:1945–1950.
23. Wang X, Chen J, Xiong H, Yu X. Genotype-phenotype associations in familial exudative vitreoretinopathy: a systematic review and meta-analysis on more than 3200 individuals. *PLoS One*. 2022;17:e0271326.

24. Kondo H. Complex genetics of familial exudative vitreoretinopathy and related pediatric retinal detachments. *Taiwan J Ophthalmol*. 2015;5:56–62.
25. Qin M, Kondo H, Tahira T, Hayashi K. Moderate reduction of Norrin signaling activity associated with the causative missense mutations identified in patients with familial exudative vitreoretinopathy. *Hum Genet*. 2008;122:615–623.
26. Kashani AH, Learned D, Nudleman E, et al. High prevalence of peripheral retinal vascular anomalies in family members of patients with familial exudative vitreoretinopathy. *Ophthalmology*. 2014;121:262–268.
27. Drenser KA, Dailey W, Vinekar A, et al. Clinical presentation and genetic correlation of patients with mutations affecting the FZD4 gene. *Arch Ophthalmol*. 2009;127:1649–1654.
28. Carrera W, Ng C, Desler C, et al. Novel FZD4 and LRP5 mutations in a small cohort of patients with familial exudative vitreoretinopathy (FEVR). *Ophthalmic Genet*. 2021;42:200–203.